

Module - IV : Analytical Techniques

Ultraviolet Spectroscopy (UV)

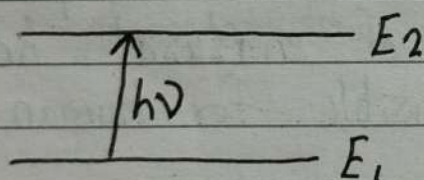
UV Region = 200 - 400 nm

[Visible region = 400 - 800 nm]

UV absorption spectroscopy arises from the transition of electrons within a molecule or ions from lower to higher energy level of the molecule or ion.

When a molecule absorb UV radiation, the electrons in the molecule undergoes transition from lower energy to higher energy. The energy difference of two states are;

$$\Delta E = E_2 - E_1 \quad \& \quad E = h\nu$$

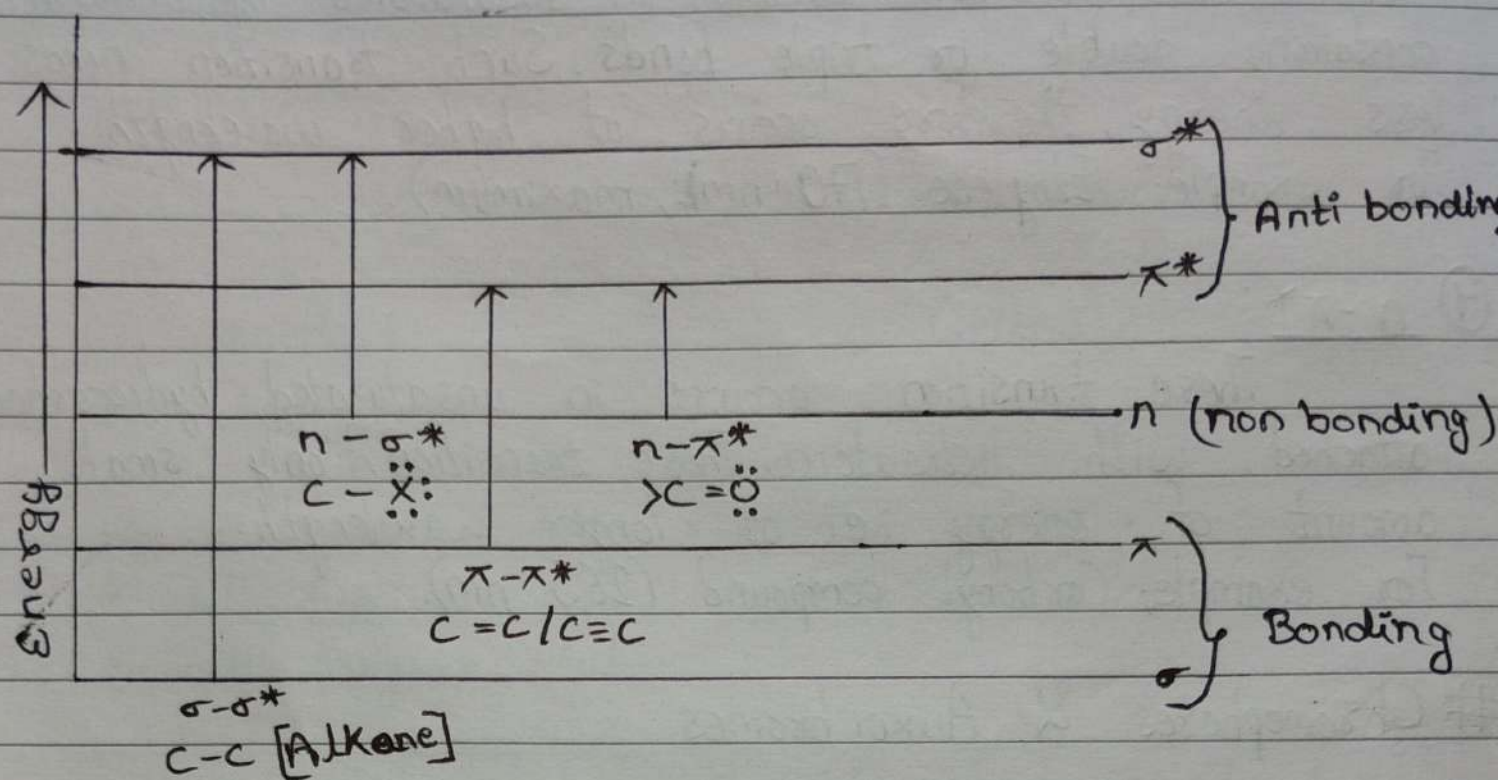


Electronic Transition

When a molecule absorb UV or visible region, the electrons are promoted from bonding to anti-bonding energy states, are called electronic transition.

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Types of Electronic Transition:



Energy Order: $\sigma \rightarrow \sigma^* > n \rightarrow \sigma^* > \pi \rightarrow \pi^* > n \rightarrow \pi^*$.

① $\sigma \rightarrow \sigma^*$:

These transition occurs only in those compound in which all the electrons are involve in single bond formation and no lone pair electrons.

Saturated hydro-carbons requires large amount of energy for σ to σ^* transition. In UV region (125-135 nm).

② $n \rightarrow \sigma^*$:

These transition occurs in saturated hydro-carbon attached by heteroatom (nitrogen, oxygen, halogen, etc) have non-bonding electrons and the absorption band appears at longer wavelength. For example; CH_3OH (183 nm).

③ $\pi-\pi^*$:

These transition occurs in unsaturated hydrocarbon containing double or triple bonds. Such transition needs less energy, therefore occurs at higher wavelength. For example; Ethylene (170 nm, maximum).

④ $n-\pi^*$:

These transition occurs in unsaturated hydrocarbon attached with heteroatom. These transition ^{requires} only small amount of energy i.e. at longer wavelength. For example; Carbonyl compound (265 nm).

Chromophores & Auxochromes

Chromophores:

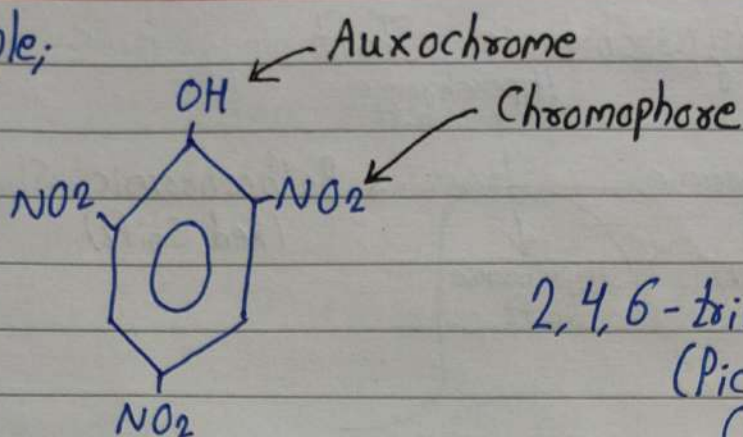
It is defined as any isolated covalently bonded group that shows a characteristic absorption and impart colour called chromophore.

A chromophore is a part of the molecule which is responsible for colour by absorbing specific wave-length of visible lights.

These are typically functional groups with double or carbonyl/nitro as a conjugated double bond.

Chlorophyll and β -carotene contain chromophore, that's why shows colour.

For example;



2,4,6-trinitro phenol (TNP)
(Picric Acid)
(Yellow)

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Types of Chromophore:

1. Chromophore which contains π -electron and they undergoes $\pi-\pi^*$ transition.

For example; Acetylene, Ethylene.

2. Chromophore which contains π -electron and non-bonding electrons, both, which causes $\pi-\pi^*$ & $n-\pi^*$ transition.

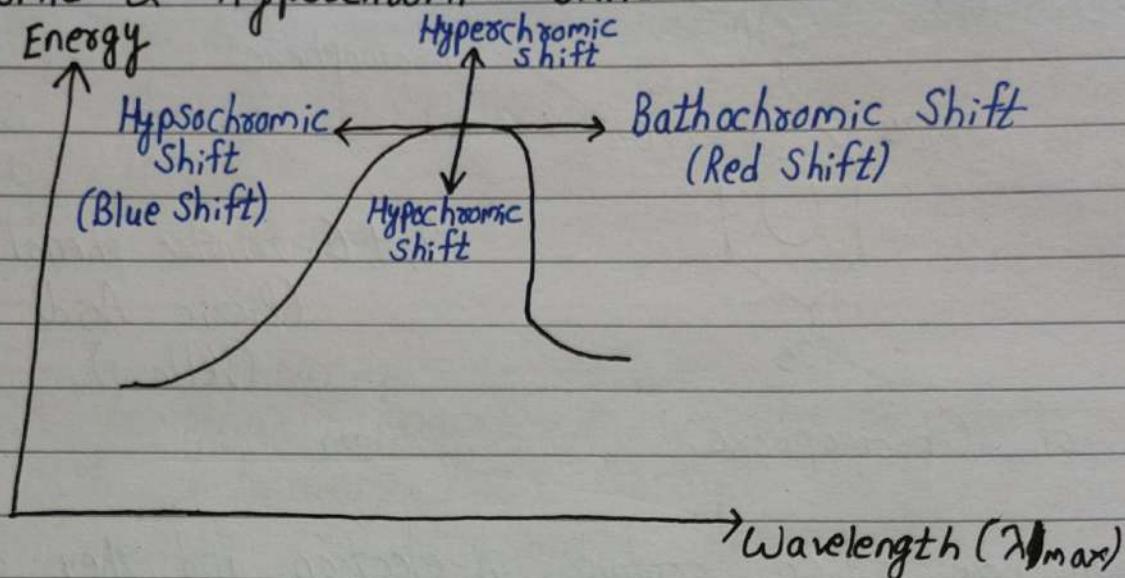
For example; Nitrile, carbonyl ($X=\ddot{O}$).
($-\underset{\text{n}}{\text{C}}\equiv\underset{\text{n}}{\text{N}}$)

Auxochromes:

Auxochrome itself does not itself act as chromophore but it is basically a colour enhancing group. It has ability to extend the conjugation of the chromophore by the sharing of non-bonding electrons - and the absorption band shifted towards longer wave-length.

For example; $-\ddot{O}H$, $-\ddot{N}H_2$, $-\ddot{O}R$, $-\ddot{N}H-R$, $-SH$, etc.

Bathochromic & Hypsochromic Shift



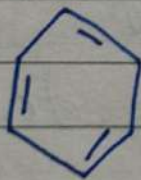
Bathochromic Shift:

When the absorption maximum shifted towards longer wavelength on certain treatment or modification, the compound is said to have bathochromic shift. It is also called red shift.

The conjugation in the molecule causes bathochromic shift.

Auxochrome can also absorb longer wavelength due to extension of conjugation.

For example;



Benzene

($\lambda_{\max} = 255 \text{ nm}$)



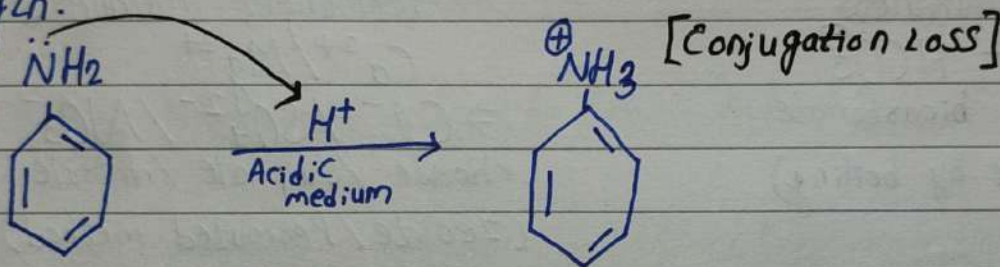
Aniline

($\lambda_{\max} = 280 \text{ nm}$)

Hypsochromic Shift:

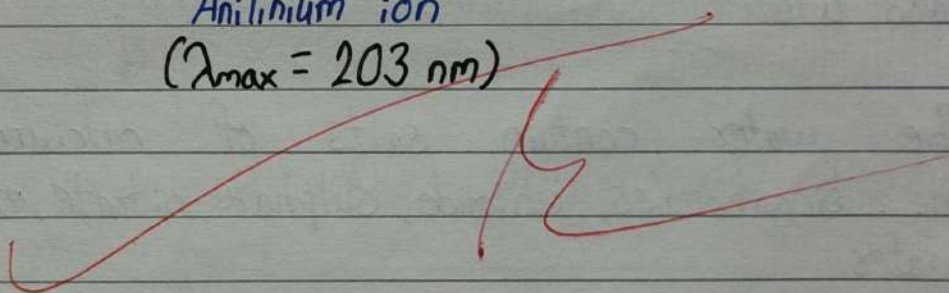
When the absorption maximum of a compound shifted towards shorter wavelength on certain treatment or modification, the compound is said to have undergone hypsochromic shift. It is also called blue shift.

For example; Aniline in acidic medium combine to form anilinium ion, at this condition the conjugation losses and the absorption maximum shifted towards shorter wavelength.



Aniline
($\lambda_{\text{max}} = 280 \text{ nm}$)

Anilinium ion
($\lambda_{\text{max}} = 203 \text{ nm}$)



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High Performance Liquid Chromatography (HPLC)

The mixture of component are separated, identify and quantify by this technique.

This is a type of chromatography in which mobile-phase is liquid. Tendency of a molecule to like or dislike water is called polarity.

If the molecule is polar and attract by water, is called hydrophilic, If the molecule is non-polar and do not attract with water, is called hydrophobic.

In HPLC, the component of a mixture are separated and purify on the basis of polarity.

There are two types:

- ① Normal Phase HPLC
- ② Reverse Phase HPLC

① Normal Phase HPLC:

The stationary phase is polar while mobile phase is non-polar.

The silica (SiO_2), polar in nature, act as stationary phase.

During normal phase HPLC, polar molecules are retained (absorb) on the stationary phase while the non-polar molecules moves fast during column is

with the mobile phase.

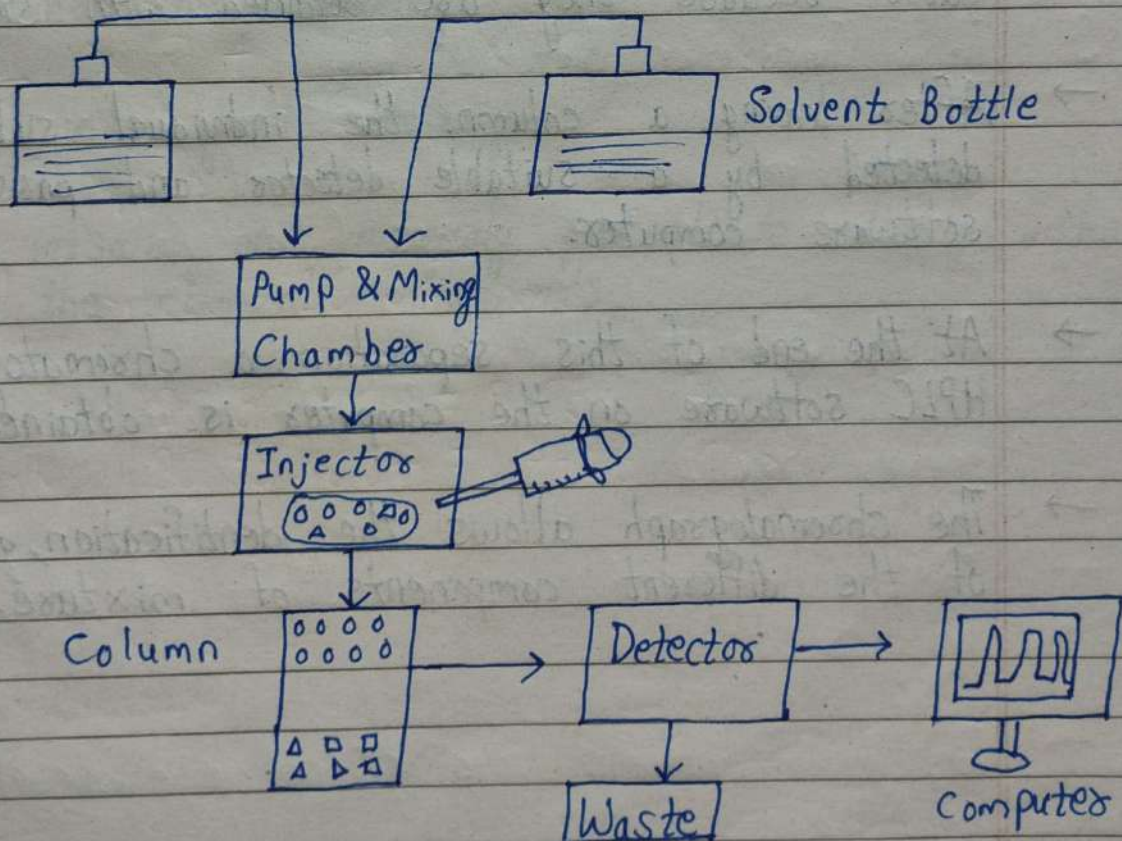
② Reverse Phase HPLC:

The stationary phase is non-polar while mobile phase is polar.

The non-polar molecules are retained (absorb) on the stationary phase while the polar molecules moves fast down in column.

In HPLC, the mixture is separated using the basic principle of ~~column~~ column chromatography, and then identify, quantify by spectrography and the solvent mixture is forced into the column at high pressure.

- Properties of HPLC:



- The purification takes place in the separation column between stationary & mobile phases
- The stationary phase is a granular mixture with very small porous particles in column.
- The mobile phase, on the other-hand, ~~is~~^{is} a solvent mixture mixed at high pressure through the separation column.
- Through a capillary (made of stainless steel), the sample is injected into the mobile phase. This sample mixture is passed to the separation column.
- Subsequently, the individual component of the sample mixture migrate through the column at different rates because they are retained with stationary phase.
- After leaving a column, the individual substances are detected by a suitable detector and passed on HPLC software computer.
- At the end of this separation, a chromatograph in the HPLC software on the computer is obtained.
- The chromatograph allows the identification, quantification of the different components of mixture.

Applications of HPLC:

- HPLC has developed into a universally applicable method, it is used in almost all areas of chemistry, bio-chemistry and pharmacy.
- Many enzymes, co-enzymes, proteins and nucleic acids are identified by HPLC.
- Ion exchanged chromatography of proteins observed in HPLC.

Advantages of HPLC:

- It has very good speed of operation.
- It has excellent efficiency.
- It has very good accuracy.
- It is versatile and extremely precise to identify and quantify the chemical components.